

Can't Beat the Clock: Evaluation of Eleven Epigenetic Clocks for the Prediction of Mortality and Length of Stay in Critically Ill ICU Patients.

David Kanter-Eivin ¹, Claudia dos Santos ², Patricia Liaw ³, Jennifer Tsang ⁴, Alexandra Binnie ⁵

1 Michael G. DeGroote School of Medicine, Faculty of Health Sciences, McMaster University, Hamilton, Canada

2 Critical Care Medicine, University of Toronto, Toronto, Canada

3 Thrombosis and Atherosclerosis Research Institute, Department of Medicine, McMaster University, Hamilton, Canada

4 Niagara Health Knowledge Institute, Niagara Health, St. Catharine's, Canada

5 Faculty of Medicine and Biomedical Sciences, University of Algarve, Faro, Portugal



17th World Congress of Intensive & Critical Care
September 16-19, 2025
Vancouver, Canada

@WCICC2025
WCICC2025.COM
#WCICC2025 #CARINGINTENSIVELY

Land Acknowledgement

The Vancouver Convention Centre is situated on the ancestral, unceded territories of the x^wməθk^wəy'əm (Musqueam), Skwxwú7mesh (Squamish), and Selíl'witulh (Tsleil-Waututh) peoples.



17th World Congress of Intensive & Critical Care
September 16-19, 2025
Vancouver, Canada

@WCICC2025
WCICC2025.COM
#WCICC2025 #CARINGINTENSIVELY

Disclosures

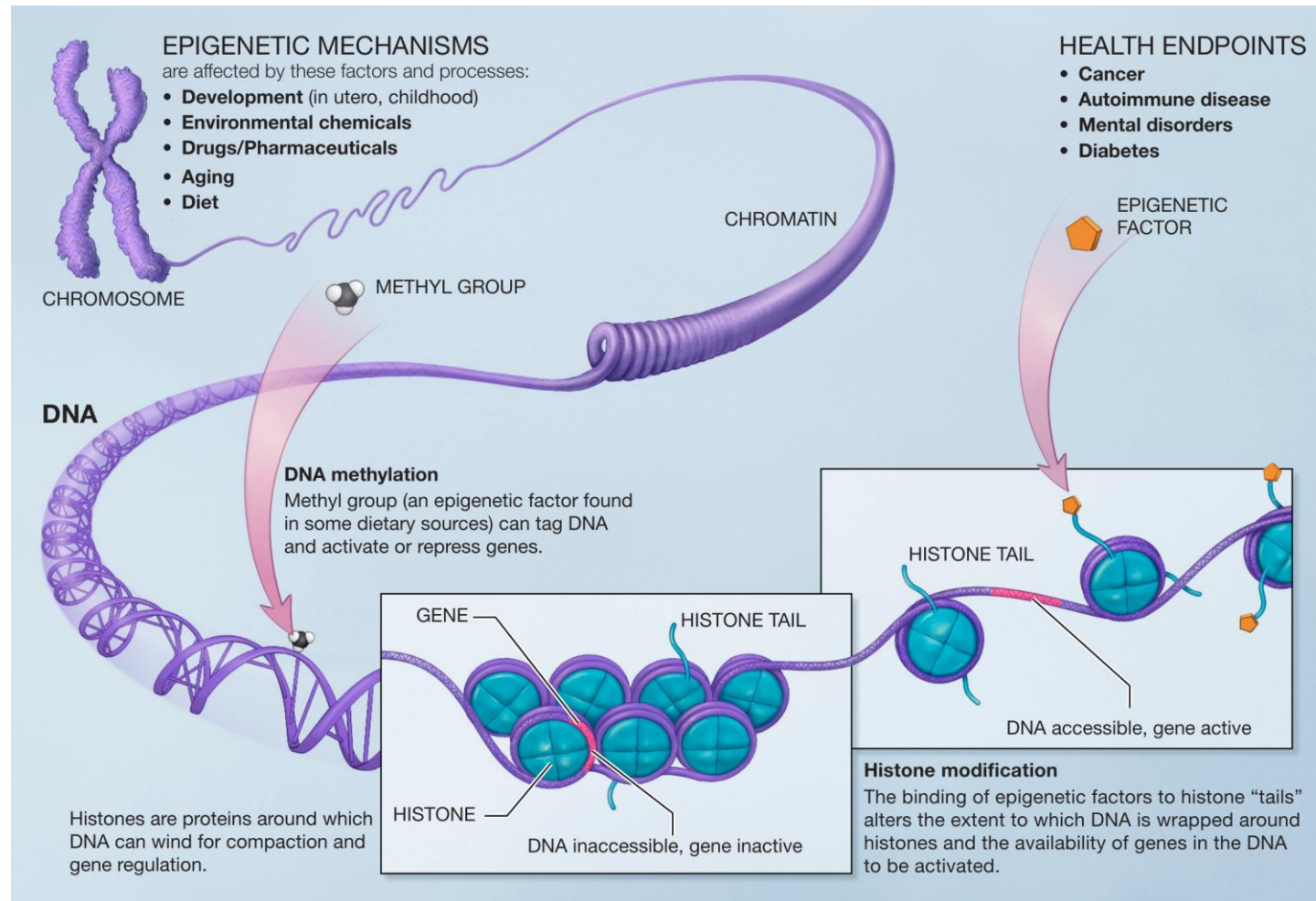
The authors do not have any conflicts of interest to disclose.

Grant Acknowledgement: We are thankful for the support of the McLaughlin Foundation Accelerator Grant in Genomic Medicine and Health Informatics (2015–2016), and Canadian Institutes of Health Research grants to Drs. dos Santos (grant number: MOP-130331) and Liaw (MOP-106503).

Availability: Slides will be made available via a QR code at the end of the talk.

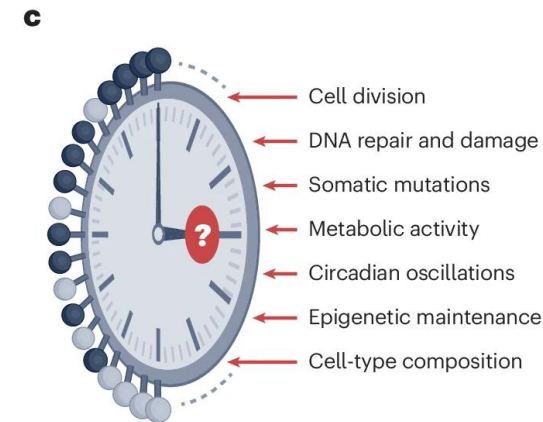
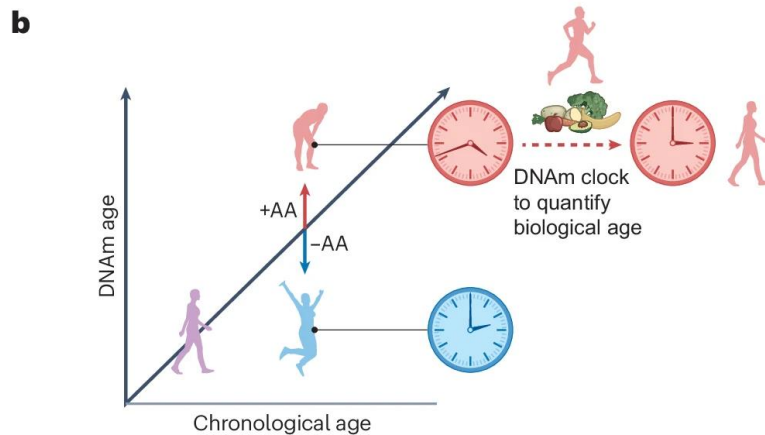
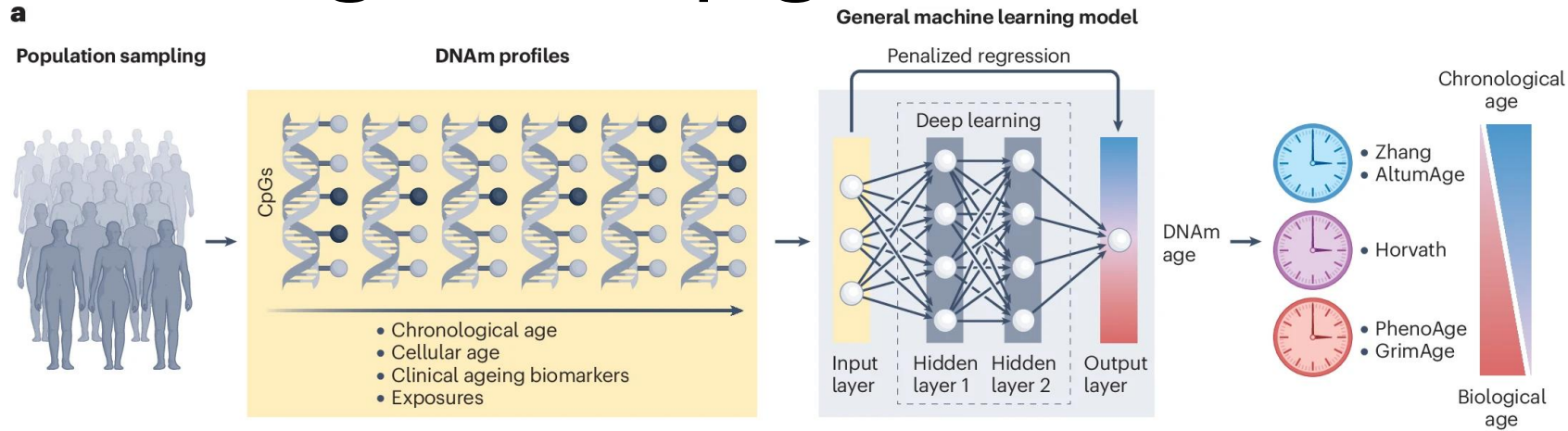


What is Epigenetics?



National Institutes of Health (2005). Online.

Epigenetic Age and Epigenetic Clocks



Teschendorff, A.E. and Horvath, S. (2025). *Nature Reviews Genetics*. 26(350-368)

The Predictive Value of Epigenetic Clocks

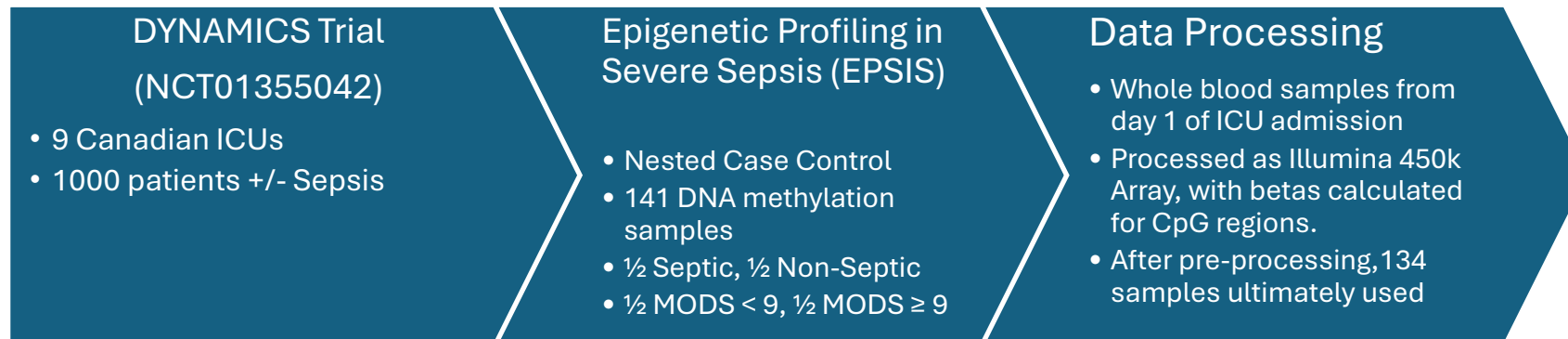
Epigenetic clocks are strongly associated with chronological age, as well as important biological and clinical outcomes:

- Chronically:
 - Lifespan, time to death, age related disease (Lu, Quach, et al., 2019; Silva et al., 2023)
 - Pro-inflammatory pathways, immuno-senescence, telomere length (Lu et al., 2019; Silva et al., 2023)
 - Physical fitness, functioning, morbidity, intrinsic capacity (Fuentelalba et al., 2025; Silva et al., 2023)
- In the acute setting:
 - Sepsis biomarkers and ICU outcomes in sepsis (Sharma-Oates et al., 2024; Binnie et al., 2020)
 - Mortality and unplanned ICU Readmission (Ho, 2023; Ho et al., 2023)



Study Design

- **Objective:** To evaluate the utility of eleven epigenetic clocks in predicting outcomes for patients admitted to ICU.
 - The **primary outcome** was **hospital mortality**.
 - The **secondary outcome** was **ICU length of stay**.
- We analyzed whole blood samples from 144 critically ill patients admitted to 4 Canadian ICUs, collected on day 1 of ICU admission.



Epigenetic Clocks

- Epigenetic ages were calculated using the dnaMethyAge R package (Wang et al., 2023).
- Age Acceleration Residual (AAR): the residual of predicted age on age acceleration, calibrated to the cohort.
 - i.e. the biological age of an individual not accounted for by their chronological age
 - **AAR generally performed similar or better to epigenetic age;** for the sake of time/understandability we omit epigenetic age analyses.

Clock Name	Citation	Trained Phenotype
PhenoAge	Levine et al. (2018)	Biological age
Multitissue Estimator	Horvath (2013)	Chronological age
Hannum Clock	Hannum et al. (2013)	Chronological age
Skin & Blood Clock	Horvath et al. (2018)	Chronological age
Zhang	Zhang et al. (2019)	Chronological age
Intrinsic Capacity Clock	Fuentealba et al. (2025)	Intrinsic capacity
DNAmTL	Lu et al. (2019)	Leukocyte telomere length
epiTOC2	Teschendorff et al. (2020)	Mitotic divisions
Mortality Risk	Zhang et al. (2017)	Mortality risk
PC GrimAge	Higgins-Chen et al. (2022)	Mortality risk
DunedinPACE	Belsky et al. (2022)	Pace of aging

Statistical Supplement

For clarity and brevity, statistical methods are abbreviated during our talk. Please see our slides for statistical methods and additional results.

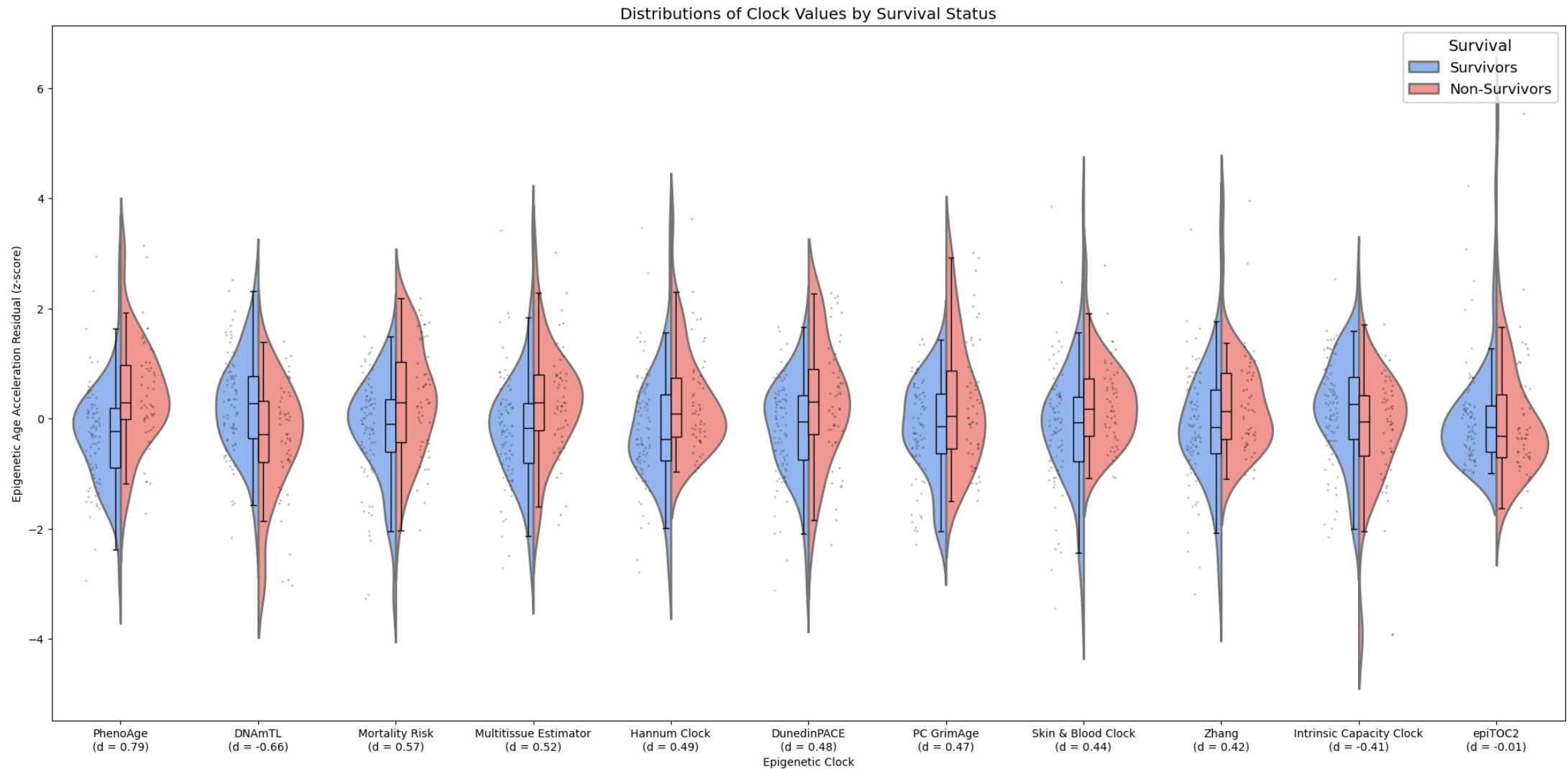
- Clock outputs were normalized by z-score for comparison purposes.
- Descriptive:
 - Correlation between epigenetic age and outcomes
 - Means across survivors/non-survivors using Cohen's d
- Survival Analysis: Covariate Adjusted Cox Proportional Hazard Models
 - Assumptions were validated using restricted cubic splines, time-interaction models, and Kaplan-Meier Curves.
- K-fold cross-validated logistic regression models were used to determine predictive performance.
- Length of Stay: Hurdle Logistic Regressions
 - Survivors' length of stay was modeled using a GLM using a Gamma distribution with log link.
 - We report multiplicative effect on LOS per 1 SD
- Analysis was performed in R version 4.4.3 and Python 3.12.11.



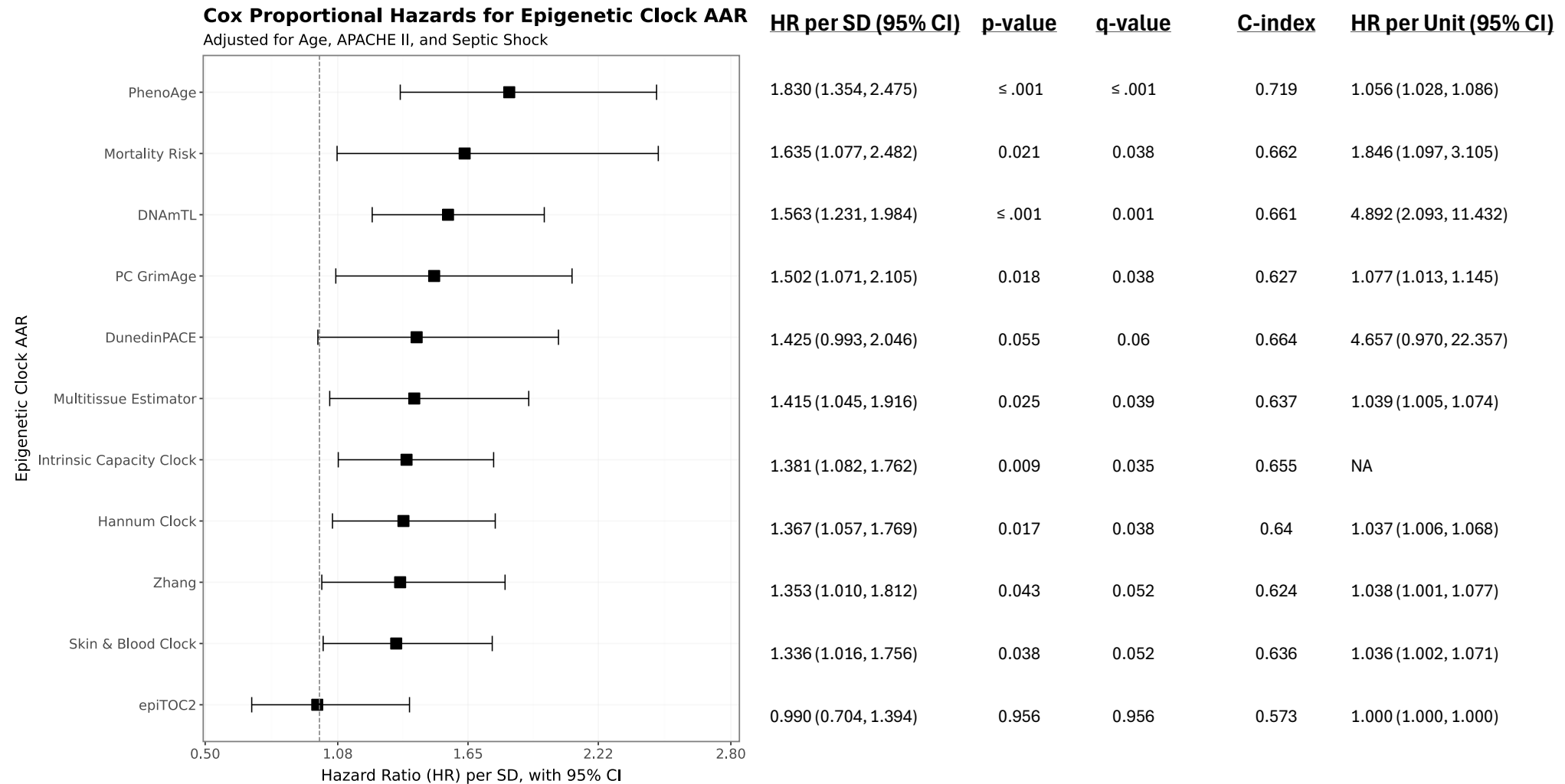
Overview

Clock	Age / Epigenetic Age					Epigenetic Age Acceleration				
	Correlation	p	95% CI	Cohen's d	p	Correlation	p	95% CI	Cohen's d	p
PhenoAge	0.308	< 0.001	0.15, 0.45	0.736	< 0.001	0.384	< 0.001	0.23, 0.52	0.791	< 0.001
Multitissue Estimator	0.222	0.010	0.05, 0.38	0.560	0.002	0.273	0.001	0.11, 0.42	0.516	0.004
DNAmTL	-0.286	0.001	-0.43, -0.12	0.723	0.000	-0.270	0.002	-0.42, -0.11	0.664	0.000
Mortality Risk	0.274	0.001	0.11, 0.42	0.616	0.001	0.246	0.004	0.08, 0.40	0.572	0.002
Hannum 2013	0.207	0.017	0.04, 0.36	0.523	0.003	0.238	0.006	0.07, 0.39	0.488	0.006
Skin & Blood Clock	0.223	0.010	0.06, 0.38	0.497	0.004	0.216	0.012	0.05, 0.37	0.436	0.009
DunedinPACE	0.229	0.008	0.06, 0.38	0.537	0.003	0.210	0.015	0.04, 0.37	0.481	0.011
Intrinsic Capacity Clock	-0.240	0.005	-0.39, -0.07	-0.502	0.005	-0.196	0.023	-0.35, -0.03	-0.406	0.029
Zhang	0.204	0.018	0.04, 0.36	0.481	0.006	0.181	0.036	0.01, 0.34	0.419	0.017
PC GrimAge	NA	NA	NA	NA	NA	0.151	0.082	-0.02, 0.31	0.470	0.014
APACHE II (DYNAMICS Cohort)	0.176	< 0.001	0.44, 0.24	NA	NA	NA	NA	NA	NA	NA
APACHE II (EPSIS Cohort)	0.131	0.132	-0.04, 0.30	0.306	0.086	NA	NA	NA	0.306	0.086
Chronological Age	0.131	0.133	-0.04, 0.29	0.296	0.086	NA	NA	NA	0.296	0.086
epiTOC2	-0.032	0.716	-0.20, 0.14	0.060	0.750	-0.059	0.501	-0.23, 0.11	0.013	0.945

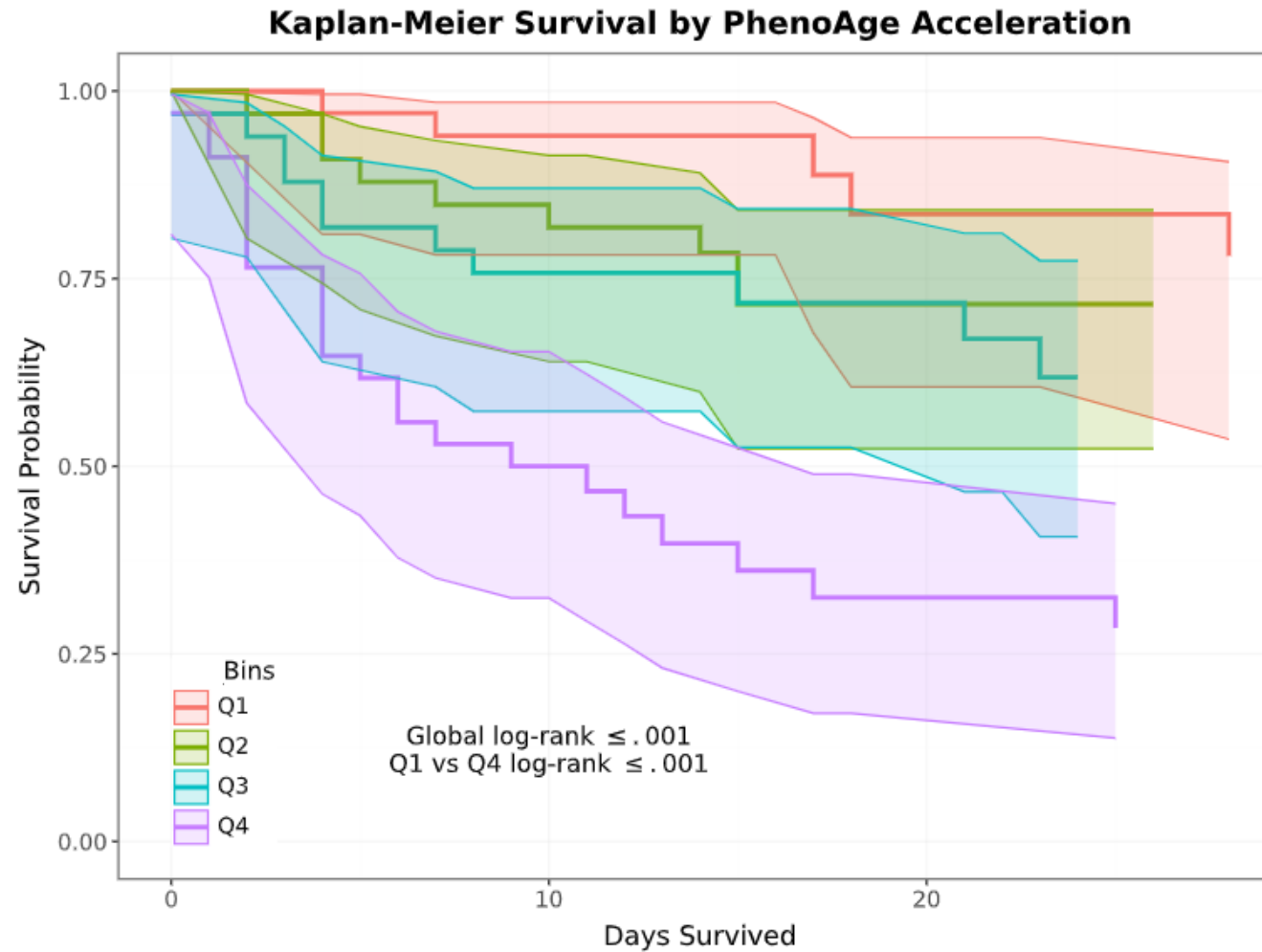
Ten clocks predicted increased epigenetic ages for non-survivors relative to survivors ($p < 0.05$), with PhenoAge demonstrating the largest effect (mean difference = 11.5 years, Cohen's $d = 0.79$).



Cox Proportional-Hazards Model



Kaplan-Meier Curve for PhenoAge AAR

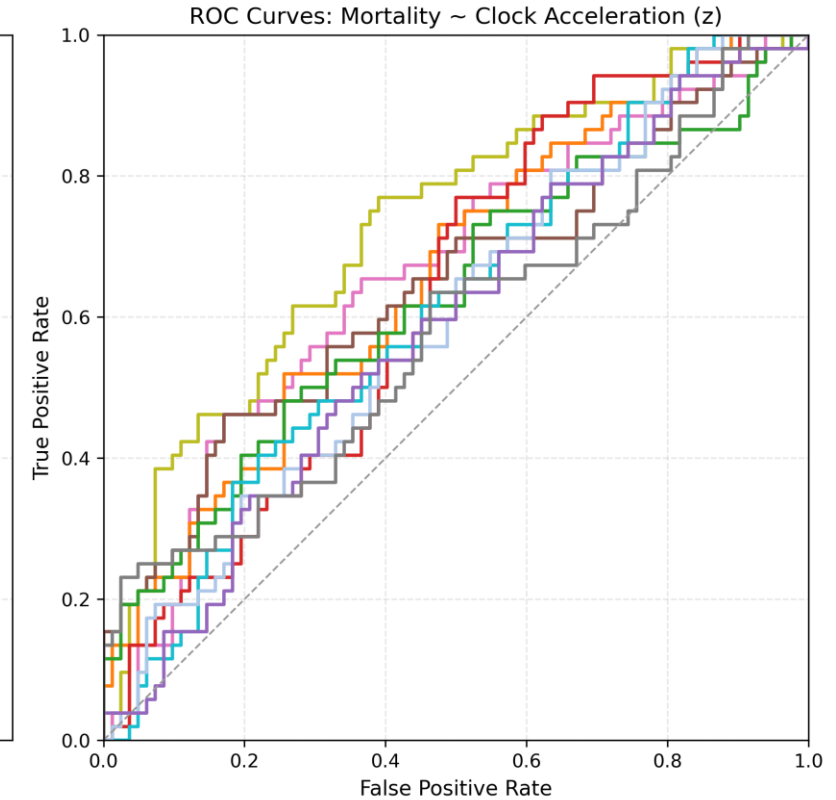
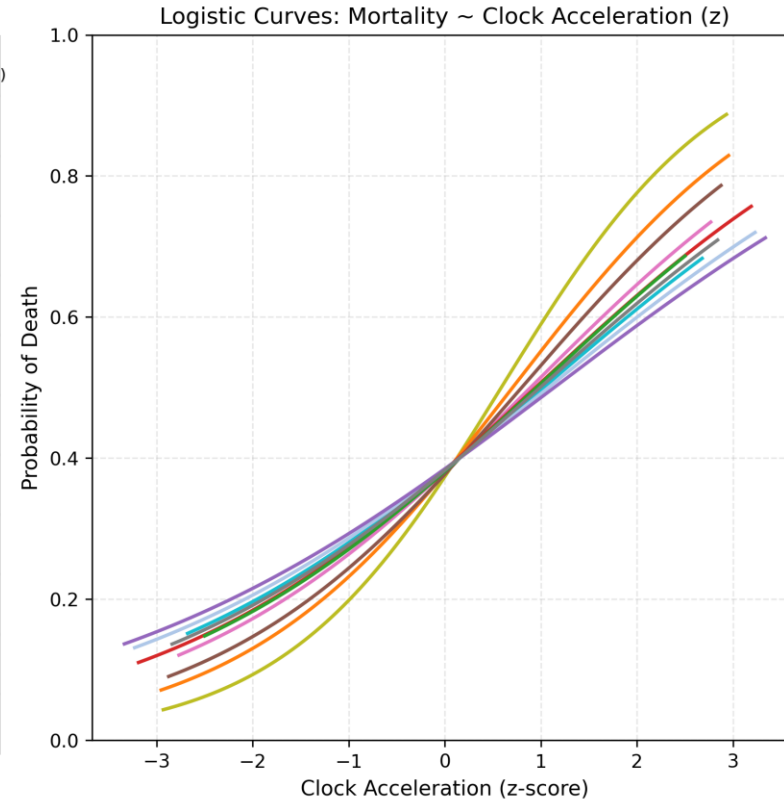


Logistic Regression on Mortality

AAR Clock Performance (AUC and OR per 1 SD)

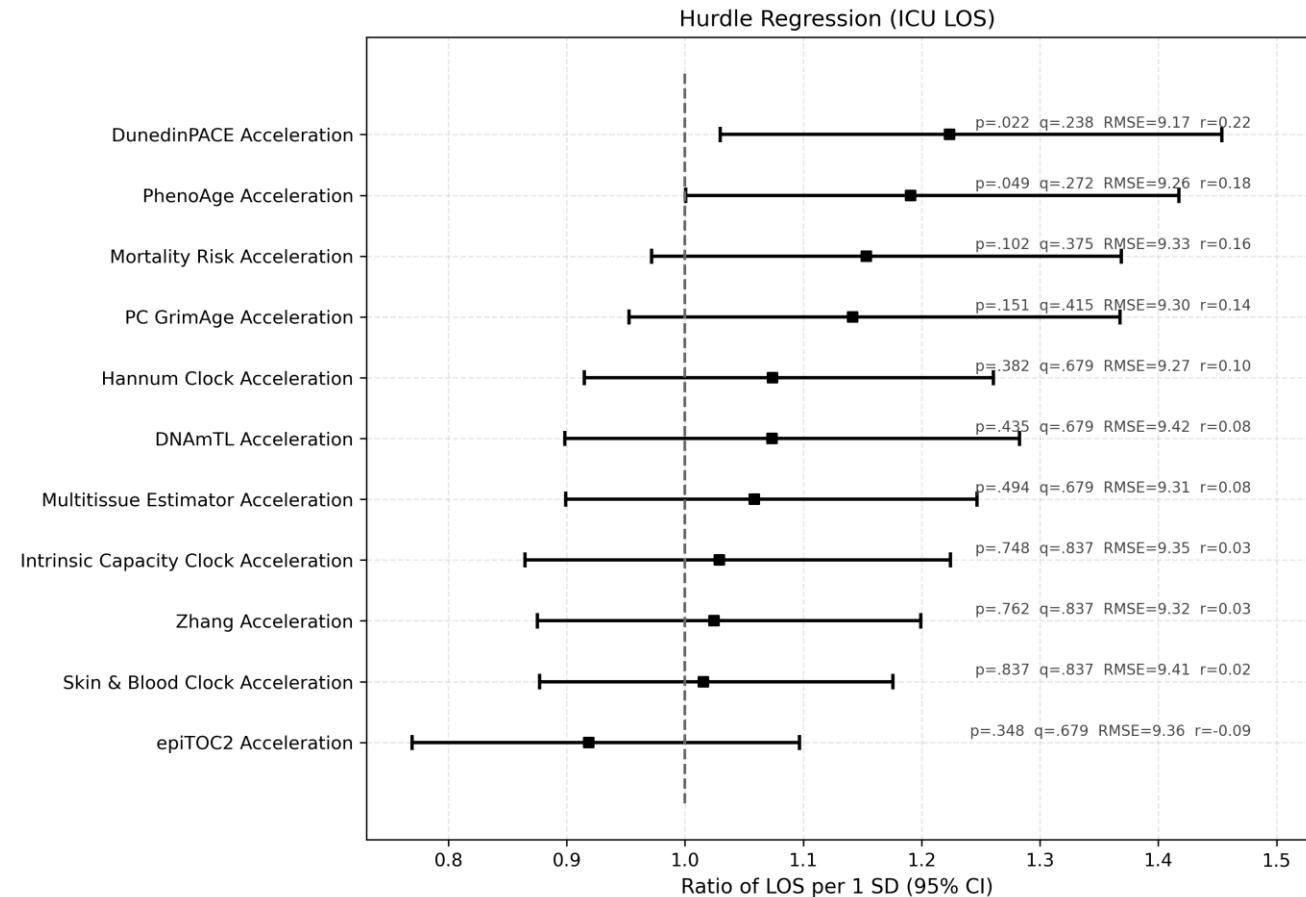
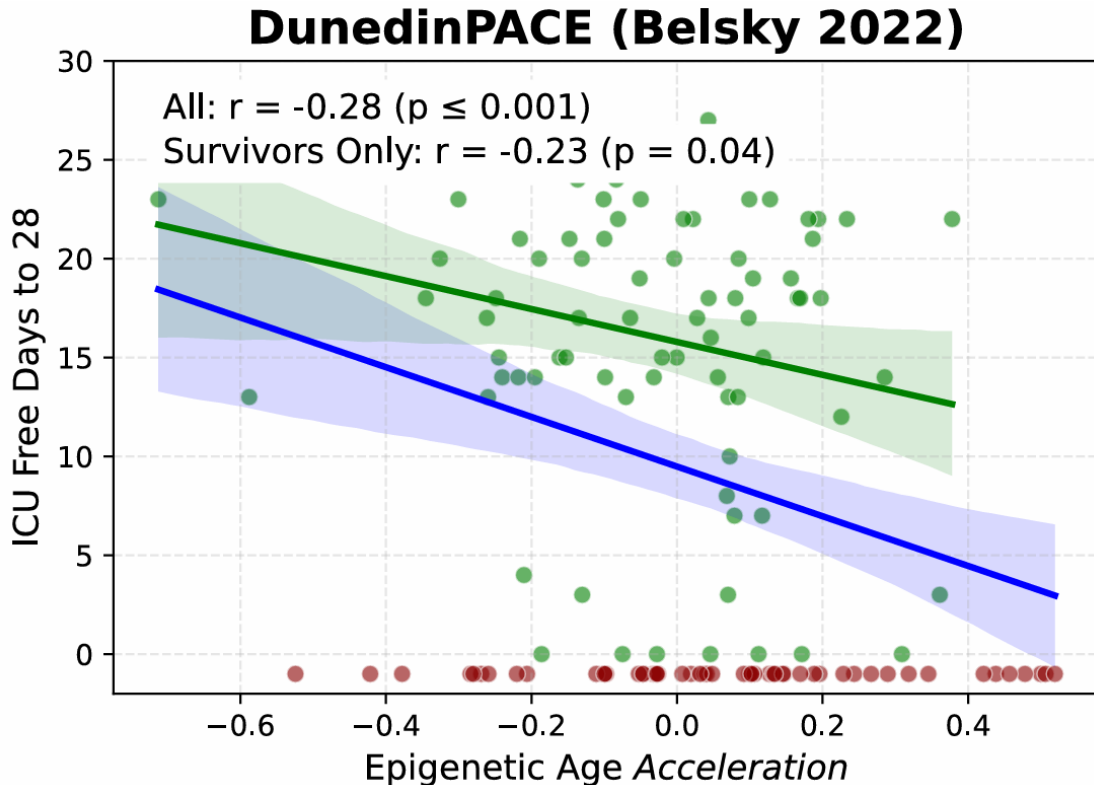
Color	Clock	AUC (95% CI)	OR per 1 SD (95% CI)	p	q	Calibration (Intercept, Slope)
	PhenoAge Acceleration	.725 (.634-.816)	2.410 (1.538-3.777)	≤ .001	.001	-.015, .960
	Multitissue Estimator Acceleration	.661 (.564-.757)	1.720 (1.165-2.540)	.006	.017	-.134, .735
	DNAmTL Acceleration	.658 (.561-.755)	2.018 (1.341-3.036)	≤ .001	.004	-.013, .971
	Mortality Risk Acceleration	.643 (.545-.740)	1.874 (1.245-2.822)	.003	.010	-.052, .895
	Hannum Clock Acceleration	.627 (.528-.725)	1.658 (1.130-2.432)	.010	.018	-.134, .703
	DunedinPACE Acceleration	.624 (.526-.723)	1.660 (1.131-2.437)	.010	.018	-.021, .952
	Skin & Blood Clock Acceleration	.609 (.510-.709)	1.592 (1.077-2.353)	.020	.027	-.168, .624
	Zhang Acceleration	.598 (.498-.697)	1.551 (1.058-2.274)	.025	.030	-.136, .697
	Intrinsic Capacity Clock Acceleration	.587 (.487-.687)	1.510 (1.044-2.183)	.028	.031	-.158, .667
	PC GrimAge Acceleration	.582 (.482-.682)	1.619 (1.113-2.355)	.012	.018	-.032, .935
	epiTOC2 Acceleration *	.465 (.365-.565)	.987 (.696-1.399)	.941	.941	-2.082, -3.549

* NS. Omitted for clarity.



Epigenetic Age and AAR for 10 out of 11 clocks outperformed age (AUC 0.568) and APACHE II (AUC 0.578).

Secondary Outcome: Length of Stay



Epigenetic Age / AAR was not convincingly associated with ICU length of stay, after controlling for mortality effects.

Limitations & Next Steps

1. Clocks are not designed to predict outcome.
 - Requires future work with larger populations to train a custom model.
 - Use of a fine-tuned GPT model (CpGPTGrimAge3) to improve task-specific performance.
2. Modest sample size limits subgroup analysis.
 - Certain clocks have differential performance in septic v.s. non-septic patients.
 - More advanced statistical methods (e.g. neural networks) are liable to overfitting.
 - Preliminary Bayesian models with PhenoAge achieved ROC of approx. 0.8.
3. Oversampling of low MODS non-survivors
 - Most clocks perform worse on the oversampled group; if anything, clock performance is underestimated. Please see supplementary materials.
4. Poorer than expected clinical score performance (e.g. APACHE- II)
 - Performance matched similar studies such as Ho et al. (2023).



Key Takeaways

Questions or Thoughts? Feel free to contact me at
David.KanterEivin@medportal.ca.



Epigenetic clocks, particularly PhenoAge, **consistently outperformed chronological age and APACHE-II score in predicting mortality** amongst critically-ill patients admitted to ICU.



Epigenetic acceleration outperformed clinical variables and comorbidities alone, suggesting that **epigenetic clocks encapsulate information not otherwise included in clinical variables.**



Discriminative ability was limited by the complexity of predicting mortality without clinical information; and the **lack of task specific training.**



Epigenetic Age / AAR was **not convincingly associated with ICU length of stay**, after controlling for mortality effects.

**SCAN HERE
FOR SLIDES**



References (Page 1)

- Belsky, D. W., Caspi, A., Corcoran, D. L., Sugden, K., Poulton, R., Arseneault, L., Baccarelli, A., Chamarti, K., Gao, X., Hannon, E., Harrington, H. L., Houts, R., Kothari, M., Kwon, D., Mill, J., Schwartz, J., Vokonas, P., Wang, C., Williams, B. S., & Moffitt, T. E. (2022). DunedinPACE, a DNA methylation biomarker of the pace of aging. *eLife*, 11, e73420. <https://doi.org/10.7554/eLife.73420>
- Binnie, A., Walsh, C. J., Hu, P., Dwivedi, D. J., Fox-Robichaud, A., Liaw, P. C., Tsang, J. L. Y., Batt, J., Carrasqueiro, G., Gupta, S., Marshall, J. C., Castelo-Branco, P., & Dos Santos, C. C. (2020). Epigenetic Profiling in Severe Sepsis: A Pilot Study of DNA Methylation Profiles in Critical Illness*. *Critical Care Medicine*, 48(2), 142–150. <https://doi.org/10.1097/ccm.0000000000004097>
- Fuentealba, M., Rouch, L., Guyonnet, S., Lemaître, J.-M., de Souto Barreto, P., Vellas, B., Andrieu, S., & Furman, D. (2025). A blood-based epigenetic clock for intrinsic capacity predicts mortality and is associated with clinical, immunological and lifestyle factors. *Nature Aging*, 5(7), 1207–1216. <https://doi.org/10.1038/s43587-025-00883-5>
- Hannum, G., Guinney, J., Zhao, L., Zhang, L., Hughes, G., Sada, S., Klotzle, B., Bibikova, M., Fan, J.-B., Gao, Y., Deconde, R., Chen, M., Rajapakse, I., Friend, S., Ideker, T., & Zhang, K. (2013). Genome-wide Methylation Profiles Reveal Quantitative Views of Human Aging Rates. *Molecular Cell*, 49(2), 359–367. <https://doi.org/10.1016/j.molcel.2012.10.016>
- Higgins-Chen, A. T., Thrush, K. L., Wang, Y., Minter, C. J., Kuo, P.-L., Wang, M., Niimi, P., Sturm, G., Lin, J., Moore, A. Z., Bandinelli, S., Vinkers, C. H., Vermetten, E., Rutten, B. P. F., Geuze, E., Okhuisen-Pfeifer, C., van der Horst, M. Z., Schreiter, S., Gutwinski, S., ... Levine, M. E. (2022). A computational solution for bolstering reliability of epigenetic clocks: Implications for clinical trials and longitudinal tracking. *Nature Aging*, 2(7), 644–661. <https://doi.org/10.1038/s43587-022-00248-2>
- Horvath, S. (2013). DNA methylation age of human tissues and cell types. *Genome Biology*, 14(10), 3156. <https://doi.org/10.1186/gb-2013-14-10-r115>
- Horvath, S., Oshima, J., Martin, G. M., Lu, A. T., Quach, A., Cohen, H., Felton, S., Matsuyama, M., Lowe, D., Kabacik, S., Wilson, J. G., Reiner, A. P., Maierhofer, A., Flunkert, J., Aviv, A., Hou, L., Baccarelli, A. A., Li, Y., Stewart, J. D., ... Raj, K. (2018). Epigenetic clock for skin and blood cells applied to Hutchinson Gilford Progeria Syndrome and *ex vivo* studies. *Aging*, 10(7), 1758–1775. <https://doi.org/10.18632/aging.101508>
- Levine, M. E., Lu, A. T., Quach, A., Chen, B. H., Assimes, T. L., Bandinelli, S., Hou, L., Baccarelli, A. A., Stewart, J. D., Li, Y., Whitsel, E. A., Wilson, J. G., Reiner, A. P., Aviv, A., Lohman, K., Liu, Y., Ferrucci, L., & Horvath, S. (2018). An epigenetic biomarker of aging for lifespan and healthspan. *Aging (Albany NY)*, 10(4), 573–591. <https://doi.org/10.18632/aging.101414>
- Lu, A. T., Quach, A., Wilson, J. G., Reiner, A. P., Aviv, A., Raj, K., Hou, L., Baccarelli, A. A., Li, Y., Stewart, J. D., Whitsel, E. A., Assimes, T. L., Ferrucci, L., & Horvath, S. (2019). DNA methylation GrimAge strongly predicts lifespan and healthspan. *Aging*, 11(2), 303–327. <https://doi.org/10.18632/aging.101684>

References (Page 2)

- Lu, A. T., Seeboth, A., Tsai, P.-C., Sun, D., Quach, A., Reiner, A. P., Kooperberg, C., Ferrucci, L., Hou, L., Baccarelli, A. A., Li, Y., Harris, S. E., Corley, J., Taylor, A., Deary, I. J., Stewart, J. D., Whitsel, E. A., Assimes, T. L., Chen, W., ... Horvath, S. (2019). DNA methylation-based estimator of telomere length. *Aging*, 11(16), 5895–5923. <https://doi.org/10.18632/aging.102173>
- Sharma-Oates, A., Sullivan, J., Pestana, D., Dos Santos, C. C., Binnie, A., & Lord, J. M. (2024). Association of Epigenetic Age and Outcome in Critically Ill Patients. *Critical Care Explorations*, 6(2), e1044. <https://doi.org/10.1097/CCE.0000000000001044>
- Silva, N., Rajado, A. T., Esteves, F., Brito, D., Apolónio, J., Roberto, V. P., Binnie, A., Araújo, I., Nóbrega, C., Bragança, J., Castelo-Branco, P., & ALFAScore Consortium. (2023). Measuring healthy ageing: Current and future tools. *Biogerontology*, 24(6), 845–866. <https://doi.org/10.1007/s10522-023-10041-2>
- Teschendorff, A. E. (2020). A comparison of epigenetic mitotic-like clocks for cancer risk prediction. *Genome Medicine*, 12(1), 56. <https://doi.org/10.1186/s13073-020-00752-3>
- Teschendorff, A. E., & Horvath, S. (2025). Epigenetic ageing clocks: Statistical methods and emerging computational challenges. *Nature Reviews Genetics*, 26(5), 350–368. <https://doi.org/10.1038/s41576-024-00807-w>
- Wang, Y., Grant, O. A., Zhai, X., McDonald-Maier, K. D., & Schalkwyk, L. C. (2023). Insights into ageing rates comparison across tissues from recalibrating cerebellum DNA methylation clock. *GeroScience*, 46(1), 39–56. <https://doi.org/10.1007/s11357-023-00871-w>
- Zhang, Q., Vallerger, C. L., Walker, R. M., Lin, T., Henders, A. K., Montgomery, G. W., He, J., Fan, D., Fowdar, J., Kennedy, M., Pitcher, T., Pearson, J., Halliday, G., Kwok, J. B., Hickie, I., Lewis, S., Anderson, T., Silburn, P. A., Mellick, G. D., ... Visscher, P. M. (2019). Improved precision of epigenetic clock estimates across tissues and its implication for biological ageing. *Genome Medicine*, 11(1), 54. <https://doi.org/10.1186/s13073-019-0667-1>
- Zhang, Y., Wilson, R., Heiss, J., Breitling, L. P., Saum, K.-U., Schöttker, B., Holleczer, B., Waldenberger, M., Peters, A., & Brenner, H. (2017). DNA methylation signatures in peripheral blood strongly predict all-cause mortality. *Nature Communications*, 8(1), 14617. <https://doi.org/10.1038/ncomms14617>
- National Institutes of Health (2005). Digital Image Accessed Online via <https://commonfund.nih.gov/sites/default/files/epigeneticmechanisms.pdf>.

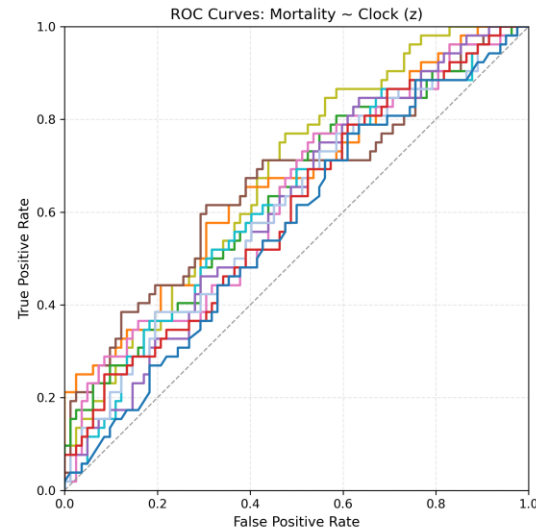
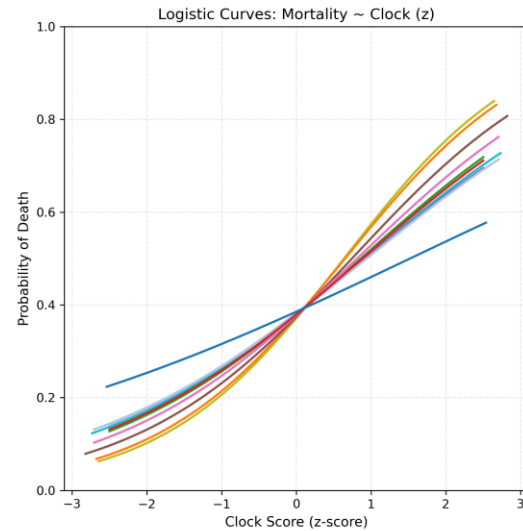


Additional Statistical Info

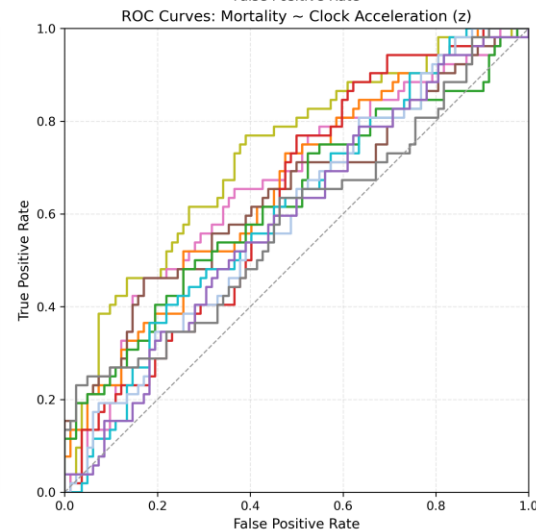
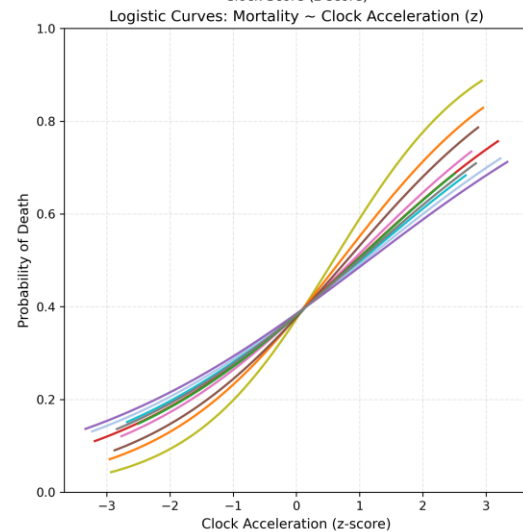
- Normality assumptions were verified using Shapiro-Wilk tests, and clocks were checked for excessive collinearity. As well, Q-Q were constructed for all clocks and residuals to validate their fit and ensure the veracity of AAR.
- q values, unless otherwise specified, refer to adjusted p values via the Benjamini-Hochberg procedure for False Discovery Rate.
- C-index (concordance index) refers to the proportion of all usable pairs whose predicted risks are correctly ordered by observed times.
- For logistic regression:
 - ORs were computed in-fold and AUC/CI was computed out of fold.
 - Area Under the Curve computed using Hanley-McNeil, with 95% CIs via Jackknife CI.
- Frequentist methods attempted (without meaningful performance gain) include ElasticNet logistic regression and splines-augmented logistic regression.
 - Feature selection followed the principles for purposeful selection [described here](#).
- Preliminary Bayesian analysis discussed uses a hierarchical logistic regression with regularized horseshoe.



Supplementary: Logistic Regression Performance for Age & Age Acceleration

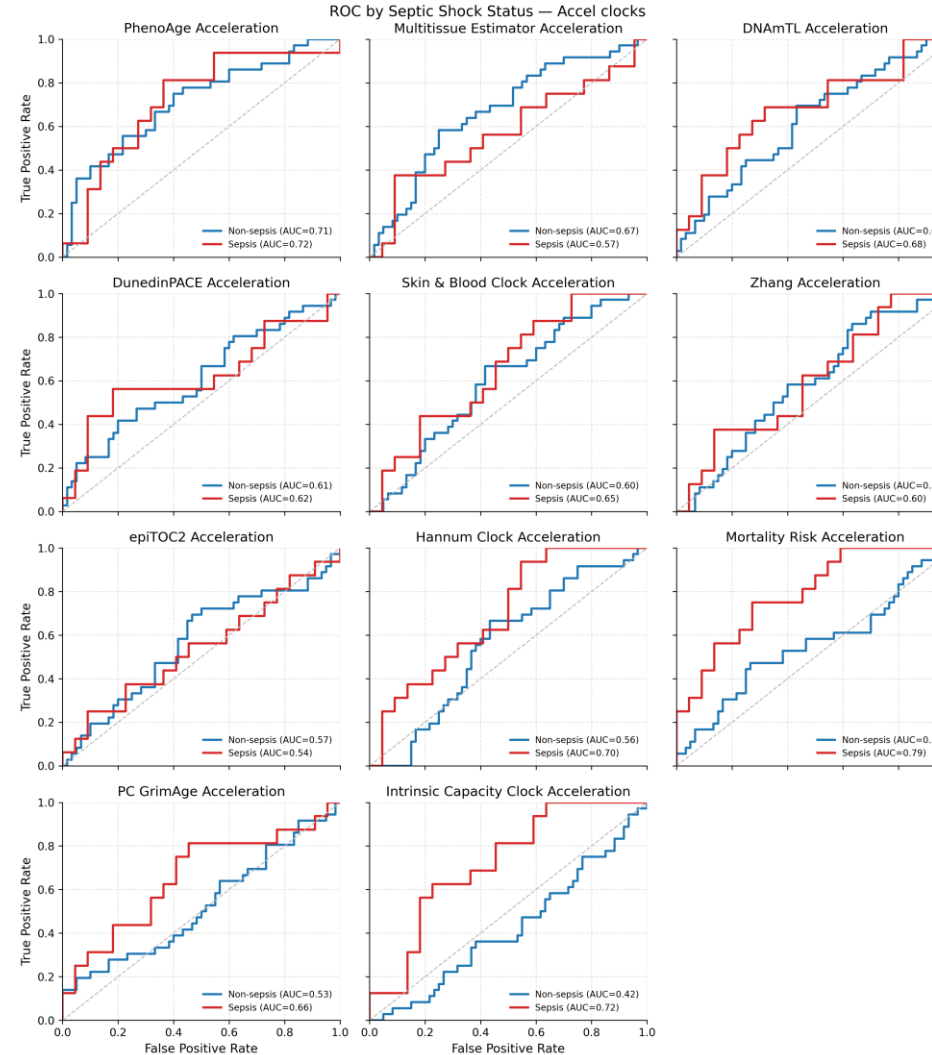


Color	Clock	AUC (95% CI)	OR per 1 SD (95% CI)	p	q	Calibration (Intercept, Slope)
Yellow	PhenoAge	.677 (.582-.773)	2.280 (1.470-3.535)	≤ .001	.002	-.016, .956
Orange	DNAmtL	.663 (.567-.760)	2.198 (1.436-3.365)	≤ .001	.002	-.012, .967
Brown	Mortality Risk	.658 (.561-.755)	1.992 (1.307-3.036)	.001	.005	-.043, .915
Green	DunedinPACE	.635 (.537-.733)	1.772 (1.197-2.623)	.004	.009	-.019, .957
Pink	Multitissue Estimator	.625 (.526-.723)	1.850 (1.227-2.788)	.003	.009	-.098, .788
Cyan	Skin & Blood Clock	.618 (.519-.717)	1.714 (1.153-2.549)	.008	.011	-.115, .726
Purple	Intrinsic Capacity Clock	.613 (.514-.712)	1.686 (1.151-2.468)	.007	.011	-.094, .790
Light Blue	Zhang	.610 (.510-.709)	1.679 (1.133-2.486)	.010	.012	-.091, .779
Red	Hannum Clock	.603 (.503-.703)	1.746 (1.181-2.582)	.005	.010	-.087, .788
Dark Blue	Age	.568 (.467-.668)	1.359 (.944-1.957)	.099	.109	-.071, .835
Light Orange	epiTOC2 *	.470 (.370-.570)	1.062 (.753-1.497)	.733	.733	-1.371, -2.001



Color	Clock	AUC (95% CI)	OR per 1 SD (95% CI)	p	q	Calibration (Intercept, Slope)
Yellow	PhenoAge Acceleration	.725 (.634-.816)	2.410 (1.538-3.777)	≤ .001	.001	-.015, .960
Pink	Multitissue Estimator Acceleration	.661 (.564-.757)	1.720 (1.165-2.540)	.006	.017	-.134, .735
Orange	DNAmtL Acceleration	.658 (.561-.755)	2.018 (1.341-3.036)	≤ .001	.004	-.013, .971
Brown	Mortality Risk Acceleration	.643 (.545-.740)	1.874 (1.245-2.822)	.003	.010	-.052, .895
Red	Hannum Clock Acceleration	.627 (.528-.725)	1.658 (1.130-2.432)	.010	.018	-.134, .703
Green	DunedinPACE Acceleration	.624 (.526-.723)	1.660 (1.131-2.437)	.010	.018	-.021, .952
Cyan	Skin & Blood Clock Acceleration	.609 (.510-.709)	1.592 (1.077-2.353)	.020	.027	-.168, .624
Light Blue	Zhang Acceleration	.598 (.498-.697)	1.551 (1.058-2.274)	.025	.030	-.136, .697
Purple	Intrinsic Capacity Clock Acceleration	.587 (.487-.687)	1.510 (1.044-2.183)	.028	.031	-.158, .667
Grey	PC GrimAge Acceleration	.582 (.482-.682)	1.619 (1.113-2.355)	.012	.018	-.032, .935
Light Orange	epiTOC2 Acceleration *	.465 (.365-.565)	.987 (.696-1.399)	.941	.941	-2.082, -3.549

Supplementary: Clock Performance, binned by Septic/Non-Septic



Supplementary: Spearman Correlation of Clocks, binned by MODS Score

